

Active-Learning Review: ~480 Candidates in 25 Minutes

Modules 06 + 07 — bypassing manual digitisation

Author: Supranab Panda · pandasupranab@gmail.com **Project:** RiceBaCI-GEE · OSF [c4mp8](#)

Version: v1.0 (2026-06-07)

What this replaces

The original v2 plan asked you to manually click 480 points across 6 GEE sessions (~4 hours). Active learning replaces that with two automated steps and one ~25-minute review session.

Step	Who does it	Time
Module 06 — generate 480 candidate points using physical rules	GEE (automated)	5 min
Module 07 — you review each candidate (Keep / Reject / Skip)	You	~25 min
Merge decisions + retrain Module 02 classifier	Agent	(later)

You already have **144 manually-drawn Fani cyclone_flood points** from Batch 17.1. These will be merged with the kept active-learning candidates — nothing wasted.

How the candidate generator works (Module 06)

It samples points that pass **strict physical thresholds**, so most candidates are already correct. You're only there to catch the ~5–10% that look ambiguous.

Cyclone-flood candidates (240 = 80 × 3 cyclones)

A pixel must satisfy **all** of: - inside the 50 km cyclone track buffer - inside ESA WorldCover 2021 cropland - Sentinel-2 **Salinity Index** > **+0.05** (salt-affected wet soil) - Sentinel-2 **NDWI** > **+0.20** (wet) - Sentinel-2 **NDVI** < **0.30** (vegetation crashed) - Sentinel-1 **VH** < **-19 dB** (smooth water) - within ±15 days of landfall

Agronomic-flood candidates (240 = 30 × 8 districts)

A pixel must satisfy **all** of: - inside ESA WorldCover 2021 cropland - JRC Global Surface Water seasonal layer ≥ 6 months/year (known transplanting flood zone) - Sentinel-2 **NDWI** > **+0.20** (wet) - Sentinel-2 **NDVI** **between 0.10 and 0.40** (young transplanted rice) - Sentinel-1 **VH**

between –22 and –16 dB (water under emerging canopy) - Sentinel-2 **SI < –0.05** (NOT salt-affected) - July–August median across non-cyclone years (2017, 2018, 2022, 2023, 2024)

These thresholds are derived from peer-reviewed SAR rice phenology literature (Bouvet et al. 2018; Pekel et al. 2016) and the Copernicus EMS surge-mapping methodology.

How the review app works (Module 07)

When you open Module 07 in the GEE Code Editor and click Run:

1. The map zooms in (scale ~1:5,000) to the first candidate point.
2. A panel on the right shows:
 - Class proposed (cyclone_flood or agronomic_flood)
 - Cyclone / event name
 - The pixel's SI, NDWI, NDVI, VH values
 - The date window
3. You look at the S2 RGB layer (default visible) and decide:

Button	When to click
✓ KEEP	The pixel looks like the proposed class (most candidates)
✗ REJECT	The pixel is obviously wrong (cloud sliver, river, urban area, unflooded fallow)
◦ SKIP	You can't tell — leave it out
4. The map auto-advances to the next candidate.	
5. You can use < Previous to backtrack if you mis-clicked.	
6. At the end, click EXPORT DECISIONS to Drive . A CSV lands in your Drive folder RiceBaCI_labels/.	

Pace

~3 seconds per candidate × 480 = **~25 minutes**. Split across 2 sessions if you prefer.

Quality target

Realistically you'll **Keep ~85–90%**, **Reject ~5–10%**, **Skip ~3–5%**. If you find yourself rejecting more than 20% in a row, send me a screenshot — it means a threshold needs tuning.

Step-by-step

Step 1 — Run Module 06 (one-time, ~5 min)

1. Open <https://code.earthengine.google.com>
2. New script → name 06_candidate_generator → paste contents of [gee/06_candidate_generator.js](#) → Save.

3. Click **Run**. The console prints candidate counts.
4. Open **Tasks** tab → click **Run** next to candidates_v1_export.
5. Wait ~5 minutes. Status turns COMPLETED.

You now have a Cloud asset at projects/durable-pulsar-486209-b5/assets/candidates_v1 containing ~480 candidates.

Step 2 — Run Module 07 (~25 min of clicking)

1. New script → name 07_active_learning_review → paste contents of gee/07_active_learning_review.js → Save.
2. Click **Run**. The first candidate appears on the map.
3. Click **Keep** / **Reject** / **Skip** for each candidate. The app auto-advances.
4. When the counter shows “All 480 candidates reviewed”, click **EXPORT DECISIONS to Drive**.
5. Open **Tasks** tab → click **Run** next to review_decisions_export.
6. The CSV lands in Drive at RiceBaCI_labels/review_decisions_2026-MM-DD.csv.

Step 3 — Send the CSV back to me

Reply in this thread with the CSV attached. I will:

- Validate the decisions (kept/reject ratio, geographic spread)
 - Merge with your 144 Fani manual cyclone_flood points
 - Retrain Module 02 classifier
 - Compute OA / F1 / UA / PA on held-out 20%
 - Re-run Modules 04 → 05 with the corrected pipeline
 - Sweep every “v2 — *pending classifier*” tag in the manuscript with real values
 - Rebuild Manuscript.docx + Supplement.docx
 - Push as Batch 17 and tag v1.0.0-rc2-real-classifier
-

What this looks like in the manuscript

The §3.4 labeling paragraph becomes:

Module 02 training labels ($n = 480$) were generated using an active-learning protocol. Candidate points were sampled from Sentinel-2 imagery using strict physical thresholds (Sentinel-1 VH < -19 dB, Sentinel-2 NDWI > +0.20, Sentinel-2 Salinity Index > +0.05 for cyclone-flood; analogous thresholds restricted to ESA WorldCover cropland and JRC Global Surface Water seasonal layer for agronomic-flood). Each candidate was then manually verified by the lead author against the Sentinel-2 RGB, false-colour, NDWI, and Sentinel-1 VH layers in the Google Earth Engine Code Editor. Of 480 candidates, K were kept, R were rejected, and S were skipped, yielding a final labeled panel of n observations (Table S10).

A new **Table S10** will document the per-cyclone keep/reject breakdown.

This narrative is **strictly stronger** than “we drew 480 polygons”: - Reproducible — anyone can re-run Module 06 and verify - Auditable — every decision is logged in the exported CSV - Threshold-justified — the physical rules cite peer-reviewed sources

Troubleshooting

Symptom	Fix
Module 06 export fails with “user memory limit”	Reduce candidatesPerCycloneClass to 60 in CFG.
Module 07 panel says “No candidates loaded”	Module 06 export task hasn’t finished yet. Check Assets tab.
Imagery in Module 07 is mostly cloudy	Cloud filter is already 60%. Some windows just have bad weather — Skip those.
You want to pause mid-review	Decisions are kept in the Code Editor session. Don’t close the tab. If you must, click EXPORT first to save progress.

End of guide.